SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Lab 7

Instructions:

Write a C++/Java program that performs the following operations using Selection Sort algorithm:

- **Functionality**: Write a function `selectionSort()` that sorts an array of integers.
- **Input**: The program should accept an unsorted array of integers.
- **Output**: Display the array after each step of the sorting process. After the sorting is complete, show the final sorted array.

Sample output:

```
Selection Sort:
Step 1: [11, 22, 45, 88, 33, 55, 19, 60]
Step 2: [11, 19, 45, 88, 33, 55, 22, 60]
Step 3: [11, 19, 22, 88, 33, 55, 45, 60]
Step 4: [11, 19, 22, 33, 88, 55, 45, 60]
Step 5: [11, 19, 22, 33, 45, 55, 88, 60]
Step 6: [11, 19, 22, 33, 45, 55, 60, 88]
Step 7: [11, 19, 22, 33, 45, 55, 60, 88]
Final Sorted Array (Selection Sort): [11, 19, 22, 33, 45, 55, 60, 88]
```